

Emmy Noether

Current English Translation Drafts

Reader note

This current working reader combines the available English translation pieces for Emmy Noether currently in hand: *Zur Invariantentheorie der Formen von n Variablen* (1910) and *Der Endlichkeitssatz der Invarianten endlicher Gruppen* (1916), the current draft and continuation on integral rational representation of invariants, and the current draft on an absolute irreducibility criterion for representations. The translations are intended as continuation bases for checking and refinement, not as proofed editions.

On the invariant theory of forms in n variables

In the projective invariant theory of forms in n variables, the principal problem has been solved: the finiteness of the system of forms has been proved. A number of further questions, however, have not been treated, namely those that concern methods for investigating the relations among forms. I should like briefly to communicate the results that I have obtained with respect to such questions, but for the sake of clarity I shall first sketch the questions in the ternary case, where they are already known.

I think first of the two fundamental theorems of the symbolic method: the theorem that all invariant formations can be represented symbolically, and the second theorem that all relations among invariants are obtained by successive application of a small number of symbolic identities; in other words, that the symbolic method gives all relations without leaving the domain of invariants.

Built upon these are the processes for generating forms: the symbolic contraction process, the non-symbolic differentiation processes, the theory of expansions into series, and so on. In this connection I should also mention a theorem proved in my dissertation: all contractions can be generated by successive application of the two possible *cogredient* contractions; by this I mean, for a symbolic product s, t, u, v , the two contractions (stu) and (vx) . Upon this theorem one can build the theory of reducentia and the reduction of systems of forms, analogously to the binary case, as I showed in my dissertation.

When we pass to the n -ary case, already the first theorem of the symbolic method holds only in a qualified sense. As is well known, for quaternary forms there already appear, besides point and plane coordinates x and u , line coordinates p_ρ , namely the minors obtained from two rows of point coordinates. Analogously, in the n -ary case we have rows of variables p_ρ whose individual elements are the minors taken from ρ rows x , and symbolic rows S_ρ whose elements behave like minors taken from ρ rows s , the latter being cogredient to the u .

The symbolic representation by determinants is known to hold only when the symbolic rows S_ρ are resolved into the rows s and these are distributed among different determinants; then only those aggregates of determinants have a real meaning that depend on the S_ρ . Which determinants depend only on the S_ρ can, to be sure, be recognized by differential equations; but in that way one obtains no survey of the totality of invariant formations, nor of the processes by which they are generated.

At this point the theorem on corresponding matrices becomes decisive. To every matrix product

$$(v^{(1)} \dots v^{(\rho)} \mid y^{(1)} \dots y^{(\rho)}),$$

where the rows of the second matrix are contragredient to those of the first, there corresponds a determinant; and conversely every determinant can be transformed into a matrix product with a prescribed number ρ of rows. Hence determinant and matrix product appear as completely equivalent, and the first theorem of the symbolic method takes the following form:

All invariant formations can be represented symbolically by matrix products.

From two symbolic rows S_ρ, T_τ one can now, with the help of rows of variables, form very easily a matrix product that contains these symbolic rows explicitly and therefore represents an invariant formation of the type $(S_\rho \mid p_\rho)(T_\tau \mid p_\tau)$, namely

$$(G_{n-\rho-\lambda} S_\rho T_\tau \mid p_{\rho+\lambda} p_{\tau-\lambda}), \quad \lambda = 1, 2, \dots, \tau \text{ or } n - \rho. \quad (1)$$

More important is the converse statement: all invariant formations can be represented explicitly by matrix products, so that the above process is the genuine extension of the binary and ternary contraction process. To prove this converse it is necessary to transfer the second theorem of the symbolic method to the n -ary case, namely to determine the totality of the independent indecomposable identities in which all rows S_ρ, p_ρ are treated as indecomposable. These identities arise from the known identities containing only rows x and u by replacing the product theorem for determinants by a system of product theorems for matrices and then combining several matrix products into a single one by means of these same product theorems. In this way one arrives at two dual formulas, of which I cite only one:

$$\sum_{i=1}^k (S_{\rho_1} \dots S_{\rho_k} \mid p_\rho x^{(i)})(S_\rho \mid x^{(i)}) = (S_{\rho_1} \dots S_{\rho_k} S_\rho \mid p_\rho). \quad (2)$$

The only specialization contained in these formulas is that a row x appears. For completely general symbolic and variable rows one reaches a *decomposition identity*, that is, an identity in which a row p_ρ appears as though resolved into the individual rows x . Clebsch already used the simplest special case of the decomposition identity in his “fundamental problem of invariant theory” in order to derive the elementary covariants occurring in series expansions.

With the aid of these identities, which mediate the passage from the non-explicit determinant expressions to matrix products, one can in fact prove that matrix representation yields the desired explicit symbolic representation.

The same formalism then gives the differentiation processes — for a row of variables gives rise to a row of symbols by replacing its entries with differential symbols — and by means of the decomposition identity it becomes possible to transfer Mertens’s proof almost literally to the n -ary case.

The finiteness theorem for the invariants of finite groups

Let the finite group G consist of the h linear transformations (with nonvanishing determinant)

$$A_1, \dots, A_h,$$

where A_k is represented by the linear transformation

$$x_\nu^{(k)} = \sum_{\mu=1}^n a_{\nu\mu}^{(k)} x_\mu \quad (\nu = 1, \dots, n),$$

or, more briefly, by

$$(x^{(k)}) = A_k(x).$$

Thus the group G carries the row (x) with elements $x_1, \dots, x_n$ into the rows $(x^{(k)})$ with elements $x_1^{(k)}, \dots, x_n^{(k)}$. Since the identity must occur among $A_1, \dots, A_h$, the row (x) itself also occurs among the rows $(x^{(k)})$.

By an *integral rational (absolute) invariant* of the group I mean an integral rational function of $x_1, \dots, x_n$ that remains identically unchanged under application of $A_1, \dots, A_h$; thus for such an invariant $f(x)$ one has

$$f(x) = f(x^{(1)}) = f(x^{(2)}) = \dots = f(x^{(h)}). \quad (1)$$

1. A full invariant system from the Galois resolvent

Formula (1) expresses that $f(x)$ is an integral rational symmetric function of the rows $(x^{(1)}), \dots, (x^{(h)})$ — in fact, since each summand $f(x)$ contains only the single row (x) , this is the simplest case, usually called the *uniform* case. By the known theorem on symmetric functions of rows, $f(x)$ can therefore be expressed integrally and rationally through the elementary symmetric functions of these rows, that is, through the coefficients $G_{\alpha_1 \dots \alpha_n}(x)$ of the “Galois resolvent”

$$\Phi(z, u) = \prod_{k=1}^h (z - (u_1 x_1^{(k)} + u_2 x_2^{(k)} + \dots + u_n x_n^{(k)})) = z^h + \sum G_{\alpha_1 \dots \alpha_n}(x) u_1^{\alpha_1} \dots u_n^{\alpha_n} z^{\alpha_0}, \quad (2)$$

where the $G_{\alpha_1 \dots \alpha_n}(x)$ are invariants of degree $\alpha_1 + \dots + \alpha_n$ in the variables x .

Thus we have proved:

Theorem. *The coefficients $G_{\alpha_1 \dots \alpha_n}(x)$ of the Galois resolvent form a complete system of invariants of the group, in such a way that every invariant can be expressed integrally and rationally in terms of these finitely many invariants.*

2. A second complete system

Starting from (1) one can also make the following still more elementary observation, which at the same time leads to a second complete system.

Suppose

$$f(x) = a + b x_\nu + \dots + c x_\nu^m + \dots,$$

where $a, b, \dots, c$ are constants. Then from (1) one has

$$h \cdot f(x) = h \cdot a + b \sum_{k=1}^h x_\nu^{(k)} + \dots + c \sum_{k=1}^h (x_\nu^{(k)})^m + \dots.$$

Hence every invariant can be assembled integrally and linearly from the special quantities

$$s_m = \sum_{k=1}^h (x_\nu^{(k)})^m \quad (m = 0, 1, 2, \dots),$$

so it is enough to prove the theorem for these special functions.

Up to a numerical factor, s_m is the coefficient of $u_1^{\alpha_1} \cdots u_n^{\alpha_n}$ in the expression

$$h \cdot s_m = \sum_{k=1}^h (u_1 x_1^{(k)} + \cdots + u_n x_n^{(k)})^m,$$

where $u = u_1 + \cdots + u_n$ is written only as a convenient abbreviation, and this expression represents the u -power sum of the h linear forms.

Now it is well known that the infinitely many power sums S_m are integral rational functions of $S_1, S_2, \dots, S_h$; their coefficients are determined by the invariants

$$S_m = \sum_{k=1}^h (x_1^{(k)} + \cdots + x_n^{(k)})^m \quad (m \leq h).$$

Thus we obtain a second complete system:

Theorem. *A complete invariant system of the group is given by all invariants $J_{\alpha_1 \dots \alpha_n}$ for which*

$$\alpha_1 + \cdots + \alpha_n \leq h,$$

where h is the order of the group.

The connection with part 1 is given by the remark that the power sums $S_1, \dots, S_h$ may be replaced by the elementary symmetric functions of $x_1^{(k)}, \dots, x_n^{(k)}$, which leads back to the complete system there given by the $G_{\alpha_1 \dots \alpha_n}(x)$.

Both results show that all invariants can be expressed integrally and rationally through those whose degree in the x does not exceed the order h of the group.

3. Rational representation

From what has just been proved one obtains consequences for rational representation. Every rational absolute invariant can — as is seen immediately by multiplying numerator and denominator by the conjugates of the denominator with respect to the group — be represented as the quotient of two integral rational absolute invariants, not necessarily relatively prime. It follows that every rational absolute invariant can be expressed rationally through the coefficients $G_{\alpha_1 \dots \alpha_n}(x)$ of the Galois resolvent $\Phi(z, u)$.

This theorem can also be proved in various easy ways without passing through the finiteness theorem; it is already formulated in Weber II, §58.

4. Final remark

Finally, I should mention that the results derived here are already implicitly contained in my paper *Körper und Systeme rationaler Funktionen*: namely, the complete system given in part 1 appears there in Theorems VI and VII, while an argument for rational representation independent of the present finiteness theorem appears there in Theorem III. Both the proof and the result become, however, considerably simpler in the special situation treated here than in the general theory.

I was led to apply those general investigations to the invariants of finite groups by conversations with E. Fischer, to whom the proof given in part 2 is due in substance as well. In the general case the complete system arising there is not rationally definable, and the remark in part 3 likewise goes back to him.

Erlangen, May 1915.

On integral rational representation of the invariants of a system of arbitrarily many basic forms

At the end of his contribution to the Schwarz Festschrift, *On the invariants of a system of arbitrarily many basic forms*, Hilbert expressed the conjecture that these invariants can be represented integrally and rationally by the finitely many invariants of the system (J, PJ) , where J denotes a complete system of invariants for the linearly independent basic forms and PJ denotes the invariants obtained from J by polar processes.

In what follows I show, in Part I, that this conjecture is indeed correct, and in fact as a simple consequence of the reduction theorem: every form in arbitrarily many rows of n variables can be represented as a sum of polars of forms that involve only n fixed rows.

In Part II I give a more conceptual proof of the reduction theorem by interpreting the question as a problem of equivalence of linear families of forms. I owe this point of view, and likewise an essential part of the reasoning employed, to Fischer's paper *On algebraic systems of modules* and to unpublished propositions of E. Fischer, whose use he kindly permitted me. Finally, in Part III I indicate how analogous considerations also lead to the most general series expansions.

I. The invariant-theoretic consequence of the reduction theorem

Let Θ be a form homogeneous in each of the $N > n$ rows

$$A_1, A_2, \dots, A_N,$$

where in general the row A_i consists of the n elements

$$a_1^{(i)}, \dots, a_n^{(i)}.$$

The coefficients of Θ may be indeterminates or quantities belonging to a given field of rationality. Let P denote an integral rational function—with rational numerical coefficients—of the polar operators

$$P_{ih} = a_1^{(i)} \frac{\partial}{\partial a_1^{(h)}} + a_2^{(i)} \frac{\partial}{\partial a_2^{(h)}} + \dots + a_n^{(i)} \frac{\partial}{\partial a_n^{(h)}},$$

where by a product such as $P_{ih}P_{jk}\Theta$ one understands the action of P_{ih} on $P_{jk}\Theta$.

Then the reduction theorem referred to above is expressed by the identity

$$\Theta = \sum P\Theta_0, \tag{1}$$

where the forms Θ_0 involve only the rows

$$A_1, A_2, \dots, A_n$$

and are obtained from Θ by polar processes P .

Now consider the basic system (F) consisting of the N forms of the same order and the same number of variables,

$$F_1, F_2, \dots, F_N,$$

whose coefficients are taken respectively to be the N rows

$$A_1, A_2, \dots, A_N.$$

Let (J) denote a complete system of invariants of

$$F_1, \dots, F_n,$$

that is, a finite set of invariants such that every invariant of

$$F_1, \dots, F_n$$

can be expressed integrally and rationally through the invariants (J) . Hilbert's conjecture to be proved is then the following: for the simultaneous invariants S of

$$F_1, \dots, F_N$$

a complete system is given by the invariants (J, PJ) .

Applying the reduction theorem (1) to a simultaneous invariant S —which can itself be regarded as a form in the N rows A_i —one sees at once that every simultaneous invariant is representable as a sum of polars of invariants that involve only the n rows

$$A_1, \dots, A_n.$$

But such forms are invariants of

$$F_1, \dots, F_n,$$

and hence can be expressed integrally and rationally through the invariants (J) . On the other hand, by the very definition of the polar processes, every form obtained from an invariant of

$$F_1, \dots, F_n$$

by a polar process is again a simultaneous invariant of

$$F_1, \dots, F_N.$$

Since the polar process preserves integral rational expressibility, it follows that every simultaneous invariant S of

$$F_1, \dots, F_N$$

can be represented integrally and rationally through the invariants (J, PJ) .

Thus we have proved:

Theorem. Every simultaneous invariant of a system of arbitrarily many basic forms is integrally and rationally expressible through the finite family (J, PJ) , where J is a complete invariant system for n linearly independent basic forms and PJ denotes all invariants obtained from J by polar operations.

This establishes Hilbert's conjecture.

On integral rational representation of the invariants of a system of arbitrarily many basic forms

II. Linear families and the proof of the reduction theorem

For the proof of the reduction theorem, we first insert some preliminaries about linear families of forms. By a *linear family of forms* over K we mean a system of forms of the same dimension so complete that, together with θ_1 and θ_2 , every combination

$$c_1\theta_1 + c_2\theta_2$$

again belongs to the system, where c_1, c_2 lie in a prescribed field K , and K is also the coefficient field of the θ . Among these forms there is always a finite number — say ϱ — that are linearly independent over K , whereas every $\varrho + 1$ forms satisfy a linear relation with coefficients in K ; the family has rank ϱ with respect to K .

We shall use the following evident fact: if L and T are two linear families over K of the same rank ϱ , and if T is a subfamily of L , then L and T are identical (equivalent).

With this understood, we return to the reduction theorem. The identity proved in Section I,

$$\theta = \sum P \cdot Z, \tag{1}$$

remains valid in analogous form if the same polar process P is applied to both sides. Thus the reduction theorem simultaneously gives a representation of *all* polars of θ by sums of the special polars PZ . On the other hand, these special polars certainly belong to the family generated by θ . The problem is therefore to prove the equivalence of suitable linear families.

For simplicity let $\theta(x, y, z)$ be of dimensions α, β, γ in the three binary rows $x_1, x_2, y_1, y_2, z_1, z_2$; the coefficients of θ are at first regarded as indeterminates a (or, if one prefers, rational numbers). We then consider the following three linear families.

1. The family L . This consists of all forms $f(x, y, z)$ obtained from θ by polar processes and having, in each of the rows x, y, z , the same dimensions as θ . In particular, L contains θ itself. If one views the $f(x, y, z)$ as forms in x, y, z and the indeterminates a , then the coefficient field is the field R of rational numbers; K must likewise be taken to be R , since only for rational c_1, c_2 does $c_1P_1 + c_2P_2$ remain a polar process. Let the rank of L over R be ϱ .

2. The family G . This consists of all forms $g(\Lambda; x, y)$ defined by

$$g(\Lambda; x, y) = f(x, y, \Lambda_1x + \Lambda_2y),$$

or, written through polar processes,

$$g(\Lambda; x, y) = \sum_{i=0}^{\gamma} \binom{\gamma}{i} \Lambda_1^{\gamma-i} \Lambda_2^i f(x, y, z)^{(i)},$$

where

$$f(x, y, z)^{(i)} = \frac{1}{i!(\gamma-i)!} \left(y \frac{\partial}{\partial z} \right)^{\gamma-i} \left(x \frac{\partial}{\partial z'} \right)^i f(x, y, z),$$

and $f(x, y, z)^{(i)}$ arises successively from $f(x, y, z)$ by the process $(yz') \frac{\partial^2}{\partial z \partial z'}$. The coefficients $g_i(x, y)$ occurring in g are therefore forms Z of the identity (1). Let G have rank δ over R .

3. The family T . This is obtained from G by the inverse polar construction by which G arose from L . Its forms $h(x, y, z)$ are defined by

$$h(x, y, z) = \sum_{i=0}^{\gamma} \binom{\gamma}{i} z_1^{\gamma-i} z_2^i g_i(x, y),$$

with

$$g_i(x, y) = \binom{\gamma}{i} f(x, y, z)^{(i)}.$$

The individual h_i , and therefore also h , are thus forms of the type PZ and sums of such forms. Let the rank of T be r .

As the construction shows, every $h(x, y, z)$ of T is obtained from θ by polar processes and has, in the rows x, y, z , the same dimensions as θ . Hence T is a subfamily of L . To conclude equivalence it remains only to prove that the ranks are equal.

Lemma a). From

$$f(x, y, \Lambda_1 x + \Lambda_2 y) = 0$$

(identically in Λ) it follows necessarily that

$$f(x, y, z) = 0.$$

This follows immediately from the identically valid relation among three binary rows

$$(xy)^2 z + (yz)^2 x + (zx)^2 y = 0,$$

which yields

$$f(x, y, (xy)^2 z + (yz)^2 x + (zx)^2 y) = 0.$$

Since $(xy)^2 = (x_1 y_2 - x_2 y_1)^2 \neq 0$, the substitution $\Lambda_1 = (yz)^2$, $\Lambda_2 = (zx)^2$ in the identity $f(x, y, \Lambda_1 x + \Lambda_2 y) = 0$ gives

$$f(x, y, z) (xy)^2 = 0,$$

hence $f(x, y, z) = 0$.

Thus there is a one-to-one correspondence between the forms of G and those of L ; moreover every linear relation in G induces, and is induced by, a linear relation in L . Therefore

$$\varrho = \delta.$$

Lemma b). From

$$h(x, y, z) = 0$$

it follows necessarily that

$$g(\Lambda; x, y) = 0.$$

Indeed, $h(x, y, z) = 0$ means that each of the terms

$$h_i(x, y) z_1^{\gamma-i} z_2^i$$

vanishes, and therefore so does every linear combination of these terms. In particular,

$$g(\Lambda; \xi, \eta) = \sum_{i=0}^{\gamma} \binom{\gamma}{i} \Lambda_1^{\gamma-i} \Lambda_2^i h_i(\xi, \eta) = 0$$

for all ξ, η , and consequently $g(\Lambda; x, y) = 0$. This proves

$$\delta = r.$$

Since $T \subseteq L$ and $\varrho = \delta = r$, the two families are equivalent. Therefore every form in L , and in particular θ itself, can be represented as a linear combination of linearly independent forms $h^{(\nu)}(x, y, z)$:

$$\theta = \sum c_{\nu} h^{(\nu)}(x, y, z) = \sum P \cdot Z.$$

Because this identity was derived for indeterminate coefficients a , it remains valid after replacing those indeterminates by elements of an arbitrary rationality field. Hence the reduction theorem is proved in the case of three binary rows.

The general proof, for N rows each of n variables, proceeds in completely parallel fashion. One again considers three linear families L , G , and T : the family L of all forms obtained from $\theta(A)$ by polar processes; the family G obtained from the forms of L by substituting

$$A_{n1} = \Lambda_1 A_{11} + \cdots + \Lambda_n A_{1n}, \quad \dots, \quad A_{nn} = \Lambda_1 A_{n1} + \cdots + \Lambda_n A_{n,n-1};$$

and the family T consisting of sums of forms PZ . The same two lemmas remain valid — in particular because among any $n + 1$ rows there is always an identically valid linear relation — and therefore L and T have the same rank. This yields once again the identity

$$\theta = \sum P \cdot Z,$$

first for indeterminate coefficients, hence also for coefficients lying in an arbitrary rationality field. Thus the reduction theorem is established in full generality.

III. General series expansions

We now indicate briefly how analogous considerations also produce the most general series expansions (for $N < n$). Let Z be a separately homogeneous form in the n rows $A_1, \dots, A_n$, each consisting of n variables, and let Δ denote the determinant of these rows. Then we first prove the identity corresponding to (1):

$$Z = \sum P \cdot H \pmod{\Delta}, \tag{2}$$

where the forms H contain only the rows $A_1, \dots, A_{n-1}$ and are derived from Z by processes P .

The proof consists in converting the relations derived in II into congruences modulo Δ . For simplicity let us again consider the case of three ternary rows x, y, z . The three linear families L, G, T are defined as before; write ϱ, δ, r for their ordinary ranks, and ϱ^*, r^* for the ranks of L and T modulo Δ .

One obtains, in place of Lemma a), the following.

Lemma a'). From

$$f(x, y, \Lambda_1 x + \Lambda_2 y) = 0$$

it follows necessarily that

$$f(x, y, z) = 0 \pmod{\Delta}.$$

Indeed, for any four ternary rows there is an identically valid relation

$$\Delta_x x - \Delta_y y + \Delta_z z = \Delta_t t,$$

and therefore, modulo Δ , always a linear relation among three rows,

$$\Lambda_x x - \Lambda_y y + \Lambda_z z \equiv 0 \pmod{\Delta}.$$

From this one deduces, exactly as before, that

$$f(x, y, \Lambda_1 x + \Lambda_2 y) \equiv 0 \pmod{\Delta} \Rightarrow f(x, y, z) \equiv 0 \pmod{\Delta}.$$

Hence $\varrho^* = \delta$.

Lemma b) from Section II remains valid without change, so $\delta = r^*$. To show in addition that $r = r^*$, Noether introduces another lemma.

Lemma c). From

$$h(x, y, z) \equiv 0 \pmod{\Delta}$$

it follows necessarily that

$$h(x, y, z) = 0.$$

The reason is that $h(x, y, z)$, being a polar form, vanishes under the familiar Ω -process. Thus one has a differential equation of the form

$$\Omega h = \sum_i \frac{\partial^2 h}{\partial x_i \partial y_i} = 0.$$

If now

$$h(x, y, z) = \Delta(x, y, z) k(x, y, z),$$

then after differentiating one obtains

$$\Omega h = (xy) k + \dots,$$

and from this it follows that h itself must vanish. Consequently

$$\varrho^* = \delta = r = r^*,$$

and because T is a subfamily of L , identity (2) is proved.

Applying identity (2) to the multiplier of Δ then leads to an expansion

$$Z = \varphi + \Delta \varphi_1 + \Delta^2 \varphi_2 + \dots + \Delta^r \varphi_r,$$

where each φ_i vanishes under the Ω -process. Repeated application of Ω and the elementary identity

$$\Omega(\Delta \psi) = \psi + \Delta \Omega(\psi)$$

then yields

$$\varphi_i = \sum P \cdot H_i,$$

and therefore the general expansion

$$Z = \sum P \cdot H + \Delta \sum P \cdot H_1 + \Delta^2 \sum P \cdot H_2 + \dots + \Delta^r \sum P \cdot H_r. \quad (3)$$

This is the most general series expansion of a form in n rows of n variables in terms of polars of forms involving only $n - 1$ rows. The corresponding expansion for forms in N rows of n variables ($N < n$) is then obtained, as Noether remarks, by a simple substitution argument.

She concludes that, together with the reductions and their combinations, these expansions exhaust the known expansion formulas for forms in arbitrarily many rows of n cogredient variables.

Erlangen, 5 January 1915.

An algebraic criterion for absolute irreducibility

A polynomial with coefficients in an arbitrarily given field is called *absolutely irreducible* if it remains irreducible in an algebraically closed field extending the coefficient field. I shall show below that, in contrast with irreducibility relative to a prescribed field, absolute irreducibility can be formulated algebraically. More precisely, the following theorem holds.

Theorem. To every polynomial in $n > 2$ variables with indeterminate coefficients one can attach a whole rational function of those coefficients and of further indeterminates — the *reducibility form* — in such a way that for every specialized polynomial of the same degree the necessary and sufficient condition for absolute irreducibility is the nonvanishing of the reducibility form. In other words: for every specialized system of coefficients for which the reducibility form is not identically zero in the auxiliary indeterminates, the polynomial is absolutely irreducible. Equivalently, the polynomial remains irreducible modulo every prime ideal of every finite algebraic field containing the coefficient field, with at most finitely many exceptional prime ideals. In particular, the containing field may be the rational numbers.

Further applications to the theory of relatively integral functions will be given elsewhere.

§1. The universal product relation

We first show that every polynomial in more than two variables with indeterminate coefficients is absolutely irreducible. Write

$$F(x) = \sum A_{i_1 \dots i_n} x_1^{i_1} \cdots x_n^{i_n}, \quad i_1 + \cdots + i_n = l, \quad (0.1)$$

$$G(x) = \sum B_{j_1 \dots j_n} x_1^{j_1} \cdots x_n^{j_n}, \quad j_1 + \cdots + j_n = m, \quad (0.2)$$

$$H(x) = F(x)G(x) = \sum C_{k_1 \dots k_n}(A, B) x_1^{k_1} \cdots x_n^{k_n}, \quad k_1 + \cdots + k_n \leq l + m. \quad (0.3)$$

Here the A and B are indeterminates, and the coefficients $C(A, B)$ are integral bilinear expressions in the A 's and B 's.

Consequently the quotients

$$D = \frac{C(A, B)}{C_{0 \dots 0}(A, B)}$$

are rational functions of the quotients $A/A_{0 \dots 0}$ and $B/B_{0 \dots 0}$. But the number of such functions is larger than the number of their arguments: namely, the three respective counts are

$$\binom{l+n}{n} - 1, \quad \binom{m+n}{n} - 1, \quad \binom{l+m+n}{n} - 1,$$

and one has the inequality

$$\binom{l+n}{n} + \binom{m+n}{n} \geq \frac{(l+m+n)!}{n!(l+m)!} = \binom{l+m+n}{n}.$$

Therefore there is at least one algebraic relation among the $D(A, B)$, and hence also among the coefficients $C(A, B)$:

$$\Phi(Z) \not\equiv 0 \quad (\text{identically in } Z), \quad \Phi(C(A, B)) = 0 \quad (\text{identically in } A, B). \quad (0.4)$$

Here Φ denotes an integral polynomial.

Hence a polynomial with indeterminate coefficients,

$$E(x) = \sum Z_{k_1 \dots k_n} x_1^{k_1} \cdots x_n^{k_n}, \quad k_1 + \cdots + k_n \leq l + m, \quad (0.5)$$

is necessarily absolutely irreducible for $n > 2$.

§2. The prime ideal of reducibility relations

The totality of all such polynomials $\Phi(Z)$ that vanish identically after the substitution $Z = C(A, B)$ forms an ideal in the polynomial ring; more precisely it forms a *prime ideal* $\mathfrak{B}$. Indeed, if a product $\Phi_1\Phi_2$ vanishes identically under the substitution, then at least one factor vanishes identically.

Since relation (0.4) holds identically in A, B , it remains valid for every specialized choice of A, B , where A, B , and therefore also $\Gamma = C(A, B)$, may be elements of an arbitrary field. Thus the coefficients Γ of every polynomial that becomes reducible in an extension field satisfy the relations

$$\Phi(\Gamma) = 0;$$

in other words, they are zeros of the prime ideal $\mathfrak{B}$, equivalently of finitely many basis polynomials for $\mathfrak{B}$. What remains is to prove the converse.

For this it is important to note that all relations among A, B, C remain valid if in the polynomials (0.1)–(0.3) we homogenize by adding a new variable y and pass to homogeneous forms $F(x, y)$, $G(x, y)$, $H(x, y)$ of dimensions $l, m, l + m$. Thus coefficients Γ may occur among the zeros of $\mathfrak{B}$ for which, say, $F(x, y)$ degenerates to a power of y ; in passing back to inhomogeneous polynomials this corresponds to a lowering of degree for $H(x)$ rather than to an actual factorization. It will be shown that, apart from this degree-lowering phenomenon, the converse always holds. In particular, for homogeneous forms the converse holds without exception, provided not all coefficients of F vanish; that is, every zero of $\mathfrak{B}$ is produced by a parameter representation

$$\Gamma = C(A, B)$$

with coefficients A, B in a suitable extension field.

Noether proves this by constructing finitely many special functions $\Phi(Z)$ directly. She associates to the polynomials (0.1)–(0.3) corresponding one-variable polynomials by means of Kronecker substitution, shows that gaps occur among the resulting exponents, and from the norms of the corresponding coefficients obtains the desired functions $\Phi(Z)$ and therefore the criterion.

§3. Kronecker substitution

Set

$$d = l + m + 1. \tag{0.6}$$

Then the Kronecker substitution associates to the polynomials F, G, H the one-variable polynomials

$$f(\xi) = \sum A_{i_1 \dots i_n} \xi^{i_1 + i_2 d + i_3 d^2 + \dots + i_n d^{n-1}}, \tag{0.7}$$

$$g(\xi) = \sum B_{j_1 \dots j_n} \xi^{j_1 + j_2 d + j_3 d^2 + \dots + j_n d^{n-1}}, \tag{0.8}$$

$$h(\xi) = \sum C_{k_1 \dots k_n}(A, B) \xi^{k_1 + k_2 d + \dots + k_n d^{n-1}}, \quad k_1 + \dots + k_n \leq l + m. \tag{0.9}$$

This correspondence is one-to-one, because from an induced exponent one can recover the exponents in the original monomial: the base- d expansion is unique, and since $d = l + m + 1$ no carrying occurs among the exponents that arise from multiplication.

Thus one has simply

$$h(\xi) = f(\xi)g(\xi), \tag{0.10}$$

and conversely, whenever a relation of the form (0.10) holds in such a way that no exponents other than the induced ones occur in f and g , one may recover the original multivariable factorization

$$H(x) = F(x)G(x). \tag{0.11}$$

Moreover, genuine gaps do appear among the induced exponents in $f(\xi)$ and $g(\xi)$. The highest exponent in $f(\xi)$ is ld^{n-1} , while the second highest is $(l-1)d^{n-1} + d^{n-2}$. Their difference is

$$d^{n-2}(d-1) > 1,$$

because $d = l + m + 1 > 3$ and $n > 2$. The same holds for $g(\xi)$.

The relations (0.10) and (0.11), first established for the indeterminates A, B , remain valid after every specialization of A, B . To preserve the degree one homogenizes (0.10) and (0.11) by means of a new variable η (and likewise y in the multivariable side). Thus a homogeneous form $H(x, y)$ factors in an algebraic extension field if and only if the corresponding binary form $h(\xi, \eta)$ splits into two factors in such a way that only the induced exponents appear in the factors.

This is the point at which Noether passes from the mere existence of algebraic relations to the construction of the reducibility form via norms. The later sections produce the explicit criterion from these one-variable relations.